

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

Let's break down the problem step by step to determine the total time it takes for Carla to download the file .

1 . **Initial Download Phase : **

- The total size of the file is 200 GB .
- Carla can download at a rate of 2 GB per minute .
- 40 % of the file is calculated as follows :

$$\begin{aligned} &0.40 \times 200 = 80 \text{ GB} \\ & \text{The time required to download 80 GB at a rate of 2 GB per minute is :} \\ & \frac{80 \text{ GB}}{2 \text{ GB/minute}} = 40 \text{ minutes} \end{aligned}$$

2 . **Windows Forced Restart : **

- The Windows update forces a restart after 40 % of the file has been downloaded .
- This restart takes 20 minutes .

3 . **Restart and Restarted Download : **

- After the restart , Carla has to start the download from the beginning .
- The entire file size is still 200 GB .
- The time required to download the full 200 GB at a rate of 2 GB per minute is :

$$\frac{200 \text{ GB}}{2 \text{ GB/minute}} = 100 \text{ minutes}$$

4 . **Total Time Calculation : **

- To find the total time taken , we sum the time for the initial download , the restart time , and the time to download the entire file again :

$$40 \text{ minutes} + 20 \text{ minutes} + 100 \text{ minutes}$$

5 . **Performing the Addition : **

- First , add the initial download time and the restart time :

$$40 + 20 = 60$$

- Then , add the time to download the entire file :

$$60 + 100 = 160$$

Therefore , the total time it takes for Carla to download the file is :

<think>
1 . Initial download : 40 minutes
2 . Windows restart : 20 minutes
3 . Full file download after restart : 100 minutes
4 . Total time : 40 + 20 + 100 = 160 minutes
</think>

<answer> 160 </answer>